

Supplementary Results on Compact Exponential Periods: Quadratic Normal Forms, Tensor-Square Residues, and Entire Exponential Integrals

Christopher D. Long Antoine-Auguste Le Blanc*

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Headlamp Software, Lexington, Kentucky, USA
galizur@gmail.com

Abstract

This note preserves three independent developments omitted from the streamlined paper *A Polynomial Kontsevich–Zagier Theorem for Compact Exponential Periods on the Affine Line*. First, for

$$H_\beta = \int_0^1 e^{\beta x^2} dx, \quad \beta \in \overline{\mathbb{Q}}^\times,$$

we prove algebraic independence of all finite families over the field generated by algebraic exponential values and derive an explicit polynomial presentation of the formal period algebra when the polynomial in the exponent has degree at most two. Second, we analyze rational horizontal endomorphisms of tensor squares of polynomial moment connections. Constancy holds through cubic polynomials; for quartics the first integral residue resonance occurs exactly for affine pure quartics, and its endomorphism algebra is computed explicitly. Third, we prove algebraic independence of finite families of values of the entire exponential integral Ein over the same exponential field. A conjugation and field-intersection argument implies that π and $\text{Ein}(1+i)$ are algebraically independent.

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1 Purpose and notation

The principal Paper III is most transparent when it is devoted solely to the all-degree formal-period theorem [7]. The results below are mathematically useful but logically independent of that proof. They are collected here so that they can be audited, cited, or separated into standalone notes without enlarging the main dependency chain.

Put $k = \overline{\mathbb{Q}}$ and

$$\mathbb{E} = k(e^\lambda : \lambda \in k), \tag{1.1}$$

*Le Blanc cryptographic author identifier: b604812530b3535b23d6dbb0f5abe32d1ed1a9de35fb4f7d04d62cff08ea19c8

the field generated by all exponential values at algebraic arguments. Any single algebraic relation over $\mathbb{E}$ involves only finitely many such exponentials, so every proof reduces to a finitely generated additive subgroup of k .

2 Differential-algebra preliminaries

2.1 E -functions and specialization

We use Beukers' refinement of the Siegel–Shidlovskii theorem [2]. In particular, at every nonsingular nonzero algebraic point, the transcendence degree of the values of an E -function vector equals the transcendence degree of the corresponding functions over $k(z)$.

Theorem 2.1 (Lindemann–Weierstrass, linear form). *If $\lambda_1, \dots, \lambda_s \in k$ are distinct, then $e^{\lambda_1}, \dots, e^{\lambda_s}$ are linearly independent over k . Consequently, if $\Gamma \subset k$ is a finitely generated additive subgroup of rank r , then*

$$\text{trdeg}_k k(e^\lambda : \lambda \in \Gamma) = r.$$

Proof. The linear assertion is classical; see [1, Chapter 1]. Choose a $\mathbb{Z}$ -basis $\omega_1, \dots, \omega_r$ of Γ . A polynomial relation among the e^{ω_j} expands into a linear relation among exponentials of distinct algebraic numbers. $\square$

2.2 Exponential fields and character extraction

Let u be a complex variable, let $d \in \{1, 2\}$, and let $\Gamma \subset \mathbb{C}$ be a finitely generated additive subgroup with basis $\omega_1, \dots, \omega_r$. Put

$$K_{\Gamma, d} = \mathbb{C}(u, e^{\omega_1 u^d}, \dots, e^{\omega_r u^d}). \quad (2.1)$$

Distinct polynomial exponentials are linearly independent over $\mathbb{C}(u)$, so this is a rational function field in the displayed exponential generators. Its constants for d/du are $\mathbb{C}$. The torus $\mathbf{T} = (\mathbb{C}^\times)^r$ acts differentially by independently rescaling the exponential generators and fixes $\mathbb{C}(u)$.

Lemma 2.2 (Extraction of one exponential character). *Suppose $g \in K_{\Gamma, d}$ and*

$$g' = \sum_{\mu \in S} h_\mu(u) e^{\mu u^d}, \quad h_\mu(u) \in \mathbb{C}(u),$$

where $S \subset \Gamma$ is finite. For every nonzero $\mu \in S$ with $h_\mu \neq 0$, there is $r_\mu(u) \in \mathbb{C}(u)$ such that

$$\frac{d}{du} (r_\mu(u) e^{\mu u^d}) = h_\mu(u) e^{\mu u^d}.$$

Proof. Work in $K_{\Gamma, d}/\mathbb{C}$. Differentiation is injective and torus equivariant. The derivative of the torus orbit of the class of g lies in the finite-dimensional span of the displayed characters; hence the orbit span of the class of g is finite-dimensional. Decompose it into torus weight spaces. A representative of the nonzero μ -weight can be chosen to be an exact semi-invariant because a torus is linearly reductive. Dividing that representative by $e^{\mu u^d}$ gives a torus-invariant element, hence an element of $\mathbb{C}(u)$. $\square$

Theorem 2.3 (Kolchin–Ostrowski). *Let K be a differential field of characteristic zero with algebraically closed constant field C . Suppose $y_1, \dots, y_m$ lie in a differential extension with the same constants and satisfy $y_j' \in K$. They are algebraically dependent over K if and only if a nonzero constant linear combination of them belongs to K .*

This is the antiderivative theorem of Kolchin and Ostrowski; see [4, 10].

Lemma 2.4 (Two rational differential equations). *Let $\alpha, \beta \in \mathbb{C}^\times$ and $c \in \mathbb{C}^\times$.*

(i) *The equation*

$$r'(u) + 2\beta ur(u) = c$$

has no solution $r \in \mathbb{C}(u)$.

(ii) *The equation*

$$r'(u) - \alpha r(u) = \frac{c}{u}$$

has no solution $r \in \mathbb{C}(u)$.

Proof. In (i), pole-order comparison excludes every finite pole, so r is a polynomial. The term $2\beta ur$ has degree $\deg r + 1$, whereas r' has degree at most $\deg r - 1$, so the sum cannot be constant. In (ii), finite poles away from zero are excluded similarly. A pole at zero makes r' have strictly larger order than both αr and c/u ; regularity at zero is also impossible because the right side has a pole. $\square$

3 Quadratic exponential integral values

For $\beta \in \mathbb{C}^\times$, define

$$H_\beta(z) = \int_0^1 e^{\beta z x^2} dx, \quad H_\beta = H_\beta(1). \quad (3.1)$$

Termwise integration and one integration by parts give

$$H_\beta(z) = \sum_{n \geq 0} \frac{\beta^n}{2n+1} \frac{z^n}{n!}, \quad 2zH'_\beta(z) + H_\beta(z) = e^{\beta z}. \quad (3.2)$$

The standard common-denominator estimate for $(2n+1)^{-1}\beta^n$ shows that H_β is an E -function when β is algebraic.

Theorem 3.1 (Functional independence of the quadratic integrals). *For distinct $\beta_1, \dots, \beta_m \in \mathbb{C}^\times$, the functions $H_{\beta_1}(z), \dots, H_{\beta_m}(z)$ are algebraically independent over the field*

$$\mathbb{C}(z, e^{\lambda z} : \lambda \in \mathbb{C}).$$

Proof. Any proposed algebraic relation uses only finitely many exponential functions. Let Γ be the additive subgroup generated by their indices and by the β_j . Put $z = w^2$ and

$$Y_\beta(w) = wH_\beta(w^2) = \int_0^w e^{\beta u^2} du.$$

Then $Y'_\beta = e^{\beta w^2}$, and the pullback of the coefficient field is contained in $K_{\Gamma,2}$.

If the Y_{β_j} were algebraically dependent, Theorem 2.3 would give constants c_j , not all zero, and $g \in K_{\Gamma,2}$ with

$$g' = \sum_j c_j e^{\beta_j w^2}.$$

For every j with $c_j \neq 0$, Lemma 2.2 produces $r_j \in \mathbb{C}(w)$ satisfying

$$r'_j + 2\beta_j w r_j = c_j,$$

contrary to Lemma 2.4(i). $\square$

Theorem 3.2 (Quadratic value independence). *For distinct $\beta_1, \dots, \beta_m \in k^\times$, the values $H_{\beta_1}, \dots, H_{\beta_m}$ are algebraically independent over $\mathbb{E}$.*

Proof. A proposed algebraic relation over $\mathbb{E}$ involves only finitely many exponential values. Let $\Gamma \subset k$ be generated by their indices and the β_j . Form the E -function vector consisting of the H_{β_j} and a basis of exponential functions for Γ . By Theorem 3.1, its functional transcendence degree is $\text{rank } \Gamma + m$. Beukers' theorem gives the same transcendence degree at $z = 1$, while Theorem 2.1 gives transcendence degree $\text{rank } \Gamma$ for the exponential-value subfield. The m quadratic values are therefore algebraically independent over that subfield. $\square$

Let $\mathcal{E}_k = k[(k, +)]$ be the group algebra with basis ε_λ . Evaluation sends ε_λ to e^λ .

Corollary 3.3 (Universal quadratic value ring). *Evaluation induces an injective homomorphism*

$$\mathcal{E}_k[T_\beta : \beta \in k^\times] \longrightarrow \mathbb{C}, \quad \varepsilon_\lambda \longmapsto e^\lambda, \quad T_\beta \longmapsto H_\beta. \quad (3.3)$$

Proof. Every polynomial involves finitely many T_β and finitely many group indices. Apply Theorem 3.2 and Lindemann–Weierstrass. $\square$

3.1 The degree-at-most-two formal algebra

Let $\mathcal{A}_{\mathbb{A}^1, \leq 2}^{\text{KZ, cpt}}$ be the subalgebra of the formal algebra in [7] generated by symbols $P(q, r, \Delta)$ with $\deg q \leq 2$ and by the symbols $E(\lambda)$. For $\beta \in k^\times$, put

$$H_\beta = P(\beta x^2, 1, [1] - [0]).$$

Lemma 3.4 (Quadratic normal form). *Let*

$$q(x) = a(x - h)^2 + c, \quad a \in k^\times, \quad h, c \in k,$$

and let $\Delta = \sum_\alpha d_\alpha [\alpha]$. Then

$$P(q, 1, \Delta) = E(c) \sum_\alpha d_\alpha (\alpha - h) H_{a(\alpha - h)^2}, \quad (3.4)$$

where the term with $\alpha = h$ is zero.

Proof. Because $\sum_\alpha d_\alpha = 0$, $\Delta = \sum_\alpha d_\alpha ([\alpha] - [h])$. On the path from h to α , substitute $x = h + (\alpha - h)t$, and then use the constant-shift identity to factor $E(c)$. $\square$

Theorem 3.5 (Explicit degree-two period algebra). *The homomorphism*

$$\mathcal{E}_k[T_\beta : \beta \in k^\times] \longrightarrow \mathcal{A}_{\mathbb{A}^1, \leq 2}^{\text{KZ, cpt}}, \quad \varepsilon_\lambda \longmapsto E(\lambda), \quad T_\beta \longmapsto H_\beta,$$

is an isomorphism, and the numerical period map on $\mathcal{A}_{\mathbb{A}^1, \leq 2}^{\text{KZ, cpt}}$ is injective.

Proof. The twisted reduction of [6] reduces a polynomial differential for a quadratic q to a constant multiple of dx plus zero-dimensional terms. Constant and linear polynomials in the exponent also reduce to zero-dimensional terms. Completing the square and applying Lemma 3.4 proves surjectivity. The composite with the numerical period map is (3.3), which is injective by Corollary 3.3; hence the displayed homomorphism and the period map are injective. $\square$

4 Tensor-square residue analysis

Let $\mathcal{M}_q$ be the homogeneous moment module associated in [5] with a polynomial q of degree $n \geq 2$. In a moment basis its connection matrix is

$$A = C + \frac{D}{z},$$

where C is semisimple,

$$\text{Spec}(D) = \left\{ -\frac{1}{n}, \dots, -\frac{n-1}{n} \right\},$$

and the compressed residue at a critical point of multiplicity m has spectrum

$$-\frac{m}{m+1}, -\frac{m-1}{m+1}, \dots, -\frac{1}{m+1}.$$

On a tensor square,

$$C^{(2)} = C \otimes I + I \otimes C, \quad D^{(2)} = D \otimes I + I \otimes D.$$

Lemma 4.1 (Compressed tensor residues). *For an eigenvalue Λ of $C^{(2)}$,*

$$\ker(C^{(2)} - \Lambda I) = \bigoplus_{\lambda+\mu=\Lambda} V_\lambda \otimes V_\mu,$$

and the compression of $D^{(2)}$ to this eigenspace is

$$\bigoplus_{\lambda+\mu=\Lambda} (\bar{D}_\lambda \otimes I + I \otimes \bar{D}_\mu).$$

Proof. Use the spectral projectors of the semisimple matrix C . Compression of $D \otimes I$ retains only the V_λ component of D in the first factor, and similarly for the second factor. Distinct ordered pairs do not mix, even when their eigenvalue sums coincide. $\square$

Theorem 4.2 (Tensor-square rigidity through degree three). *Let M be a finite direct sum of moment modules attached to polynomials of degree two or three. Every rational horizontal endomorphism of $M^{\otimes 2}$ and of $\text{Sym}^2 M$ is constant in the tensor moment basis.*

Proof. The possible residue eigenvalues of M are $-1/2, -1/3, -2/3$. Those of $D^{(2)}$ lie in $[-4/3, -2/3]$, an interval of width $2/3$. A pole at zero of order $m \geq 1$ would make $-m$ an eigenvalue of $\text{ad } D^{(2)}$, which is impossible; poles away from zero are excluded by order comparison. Thus a rational horizontal endomorphism is polynomial.

If its degree at infinity were $d \geq 1$, compression of the two leading coefficient equations to an eigenspace of $C^{(2)}$ would make d an eigenvalue of the commutator with the compressed tensor residue. By Lemma 4.1, that residue again has spectrum in an interval of width $2/3$. Hence d cannot be a positive integer. The symmetric square is a horizontal direct summand of the tensor square. $\square$

4.1 The first quartic resonance

After an algebraic affine change of the source, a quartic has the form

$$q(x) = \frac{x^4}{4} + ax^2 + bx + c. \quad (4.1)$$

It is affine-equivalent to a pure quartic $\alpha(x-h)^4 + \beta$ exactly when $a = b = 0$ in this normalization.

We use the row-module convention, so a horizontal endomorphism Q satisfies

$$Q' = [Q, C + D/z].$$

In the basis

$$v_0 = 1, \quad v_1 = x, \quad v_2 = x^2 + a,$$

the row-connection matrices are

$$D = \text{diag} \left(-\frac{1}{4}, -\frac{1}{2}, -\frac{3}{4} \right), \quad (4.2)$$

and

$$C = \begin{pmatrix} c - \frac{a^2}{2} & -\frac{5ab}{4} & \frac{a^3}{2} - \frac{3b^2}{4} \\ \frac{3b}{4} & c - a^2 & -\frac{5ab}{4} \\ \frac{a}{2} & \frac{3b}{4} & c - \frac{a^2}{2} \end{pmatrix}.$$

These formulas follow by dividing qv_j by q' and taking the remainder. On $\text{Sym}^2 V$, in the ordered product basis

$$v_0^2, v_0v_1, v_0v_2, v_1^2, v_1v_2, v_2^2,$$

one has

$$D^{(2)} = \text{diag} \left(-\frac{1}{2}, -\frac{3}{4}, -1, -1, -\frac{5}{4}, -\frac{3}{2} \right). \quad (4.3)$$

Lemma 4.3 (Pole exclusion for a non-pure quartic). *If $(a, b) \neq (0, 0)$, a rational horizontal endomorphism of the tensor square has no pole at zero.*

Proof. The width of the spectrum in (4.3) is one. A pole can only have order one and, after projection to the symmetric square, its leading term is the map $E_{0,5}$ from the lowest residue line to the highest. Write

$$Q = z^{-1}E_{0,5} + Q_0.$$

The coefficient equations are

$$[Q_0, D^{(2)}] = -[E_{0,5}, C^{(2)}], \quad [Q_0, C^{(2)}] = 0.$$

Solving the first equation entrywise gives

$$Q_0 = \begin{pmatrix} u_0 & 0 & a & 0 & b & 0 \\ 0 & u_1 & 0 & 0 & 0 & -2b \\ 0 & 0 & u_2 & u_3 & 0 & -2a \\ 0 & 0 & u_4 & u_5 & 0 & 0 \\ 0 & 0 & 0 & 0 & u_6 & 0 \\ 0 & 0 & 0 & 0 & 0 & u_7 \end{pmatrix}.$$

The $(0, 0)$ entry of the second equation is a^2 , so $a = 0$. If $b \neq 0$, four further entries give

$$u_1 = u_0, \quad u_0 - u_2 = 1, \quad u_3 = -2, \quad u_1 - u_2 - u_3 = 0,$$

a contradiction. Hence $b = 0$ as well. □

Theorem 4.4 (Quartic tensor-square classification). *Let q be a quartic polynomial.*

- (i) *If q is not affine-equivalent to a pure quartic, every rational horizontal endomorphism of $\mathcal{M}_q^{\otimes 2}$ is constant.*
- (ii) *If q is pure, then*

$$\begin{aligned}\mathrm{End}^\nabla(\mathcal{M}_q^{\otimes 2}) &\simeq M_2(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus M_3(\mathbb{C}) \oplus M_2(\mathbb{C}), \\ \mathrm{End}^\nabla(\mathrm{Sym}^2 \mathcal{M}_q) &\simeq M_2(\mathbb{C}) \oplus \mathbb{C} \oplus M_2(\mathbb{C}) \oplus \mathbb{C}.\end{aligned}$$

The nonconstant entries are the factors z and z^{-1} joining the two extreme residue lines.

Proof. If q is not pure, its critical multiplicities are $1 + 1 + 1$ or $2 + 1$. The compressed tensor residues then lie in an interval of width less than one, so positive degree at infinity is excluded exactly as in Theorem 4.2; Lemma 4.3 excludes the only possible pole at zero.

For a pure quartic, C is scalar. The tensor residue weights are

weight	$-\frac{1}{2}$	$-\frac{3}{4}$	-1	$-\frac{5}{4}$	$-\frac{3}{2}$
multiplicity	1	2	3	2	1.

An entry joining weights d_i, d_j is a constant multiple of $z^{d_j - d_i}$ and is rational exactly when the difference is integral. Grouping weights modulo $\mathbb{Z}$ gives block dimensions 2, 2, 3, 2 on the full tensor square. On the symmetric square the multiplicities are 1, 1, 2, 1, 1, giving block dimensions 2, 1, 2, 1. $\square$

Remark 4.5. The quartic endomorphisms are symmetries of a homogeneous tensor connection, not polynomial equations for the inhomogeneous moment orbit. The affine linearity theorem in [7] proves that they create no new polynomial relations among period values.

5 Entire exponential-integral values

Define

$$\mathrm{Ein}(z) = \int_0^z \frac{1 - e^{-t}}{t} dt = \sum_{n \geq 1} \frac{(-1)^{n-1} z^n}{n n!}. \quad (5.1)$$

For $\alpha \in \mathbb{C}^\times$, put $E_\alpha(z) = \mathrm{Ein}(\alpha z)$. Then

$$z E'_\alpha(z) = 1 - e^{-\alpha z}. \quad (5.2)$$

When α is algebraic, the usual denominator estimate for α^n/n and (5.2) show that E_α is an E -function.

Theorem 5.1 (Functional independence of entire exponential integrals). *For distinct $\alpha_1, \dots, \alpha_m \in \mathbb{C}^\times$, the functions $E_{\alpha_1}(z), \dots, E_{\alpha_m}(z)$ are algebraically independent over*

$$\mathbb{C}(z, e^{\lambda z} : \lambda \in \mathbb{C}).$$

Proof. Any proposed relation uses only finitely many exponentials; let Γ be the subgroup generated by their indices and the α_j . If the displayed functions were algebraically dependent over $K_{\Gamma,1}$, Theorem 2.3 would give constants c_j , not all zero, and $g \in K_{\Gamma,1}$ such that

$$g' = \frac{\sum_j c_j}{z} - \sum_j \frac{c_j}{z} e^{-\alpha_j z}.$$

For any j with $c_j \neq 0$, character extraction gives $r_j \in \mathbb{C}(z)$ satisfying

$$r'_j - \alpha_j r_j = -\frac{c_j}{z},$$

contrary to Lemma 2.4(ii). $\square$

Theorem 5.2 (Algebraic independence of Ein-values). *For distinct $\alpha_1, \dots, \alpha_m \in k^\times$, the values*

$$\text{Ein}(\alpha_1), \dots, \text{Ein}(\alpha_m)$$

are algebraically independent over $\mathbb{E}$.

Proof. A relation over $\mathbb{E}$ involves finitely many exponential values. Form the E -function vector consisting of the E_{α_j} , the constant function 1, and exponential functions whose indices generate the finite coefficient field. Apply Theorem 5.1, Beukers' transcendence- degree equality at $z = 1$, and Theorem 2.1. $\square$

The result is compatible with the classical first-order theory of E -functions [9] and with the structural work of Rivoal and Roques [8]. The effective computation of relation ideals for a fixed tuple is treated by Fischler and Rivoal [3].

6 Conjugation and an ordinary period

Lemma 6.1 (Intersection of one-variable algebraic closures). *Let k be algebraically closed, let x, y be algebraically independent over k , and let t lie in a common overfield. If t is algebraic over both $k(x)$ and $k(y)$, then $t \in k$.*

Proof. If $t \notin k$, then t is transcendental over k . Algebraic dependence makes both x and y algebraic over $k(t)$, so $k(x, y, t)$ has transcendence degree one. This contradicts the inclusion of $k(x, y)$, which has transcendence degree two. $\square$

Theorem 6.2 (Conjugate-pair principle). *Let $F(z) = \sum_{n \geq 0} a_n z^n$ have real algebraic coefficients and converge at α and $\bar{\alpha}$. If $F(\alpha), F(\bar{\alpha})$ are algebraically independent over k , then every real transcendental number τ is algebraically independent from $F(\alpha)$ over k .*

Proof. If τ and $F(\alpha)$ were algebraically dependent, then τ would be algebraic over $k(F(\alpha))$. Complex conjugation fixes τ , preserves k , and sends $F(\alpha)$ to $F(\bar{\alpha})$, so τ would also be algebraic over $k(F(\bar{\alpha}))$. Apply Lemma 6.1. $\square$

Theorem 6.3 (Algebraic independence of π and $\text{Ein}(1+i)$). *The numbers π and $\text{Ein}(1+i)$ are algebraically independent over $\bar{\mathbb{Q}}$.*

Proof. By Theorem 5.2, the conjugate pair $\text{Ein}(1+i), \text{Ein}(1-i)$ is algebraically independent over k . The Taylor coefficients of Ein are real, and π is real and transcendental. Apply Theorem 6.2. $\square$

The second number is itself a compact two-dimensional exponential period:

$$\text{Ein}(\alpha) = \alpha \int_0^1 \int_0^1 e^{-\alpha xy} dy dx. \quad (6.1)$$

Thus Theorem 6.3 is a mixed algebraic-independence result between an ordinary period and a compact exponential period. It does not assert a formal-period theorem for the category generated by all ordinary and exponential periods.

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